

# Functions of Several Variables

## Chapter 4 (Part 1): Functions of More than One Variable, Domains & Level Curves

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# Motivation: Quantities with Multiple Inputs (Slide 3)

In real-world applications, many objects and quantities are determined by **more than one independent variable**:

## Geometric Examples

- **Area of a rectangle:**

$$A = xy$$

( $x$  is length,  $y$  is width).

- **Volume of a circular cone:**

$$V = \frac{1}{3}\pi r^2 h$$

( $r$  is radius,  $h$  is height).

## Physical Examples

- **Kinetic energy:**

$$E_k = \frac{1}{2}mv^2$$

( $m$  is mass,  $v$  is velocity).

- **Potential energy:**

$$E_p = mgh$$

( $m$  is mass,  $h$  is height).

# Functions of Two Variables & Natural Domain (Slide 4)

## Definition (Function of Two Variables)

Let  $z = f(x, y)$  be a function with two variables  $x$  and  $y$ .

- $x$  and  $y$  are the **independent variables**.
- $z$  is the **dependent variable**.
- Unless otherwise specified, the **domain**  $D$  is the **largest subset of  $\mathbb{R}^2$**  on which  $f$  is mathematically well-defined.

## Boundary Convention for Domains in $\mathbb{R}^2$

- **Strict inequality ( $>$  or  $<$ ):** Boundary drawn as a **dashed curve** (points excluded).
- **Non-strict inequality ( $\geq$  or  $\leq$ ):** Boundary drawn as a **solid curve** (points included).

# Domain Example 1: Parabola and Line Restrictions (Slide 4)

## Example (MA1521 Slide 4)

Find and sketch the domain of  $f(x, y) = \frac{x + 2y}{\sqrt{x - y^2}} + \ln(8 - x - y)$ .

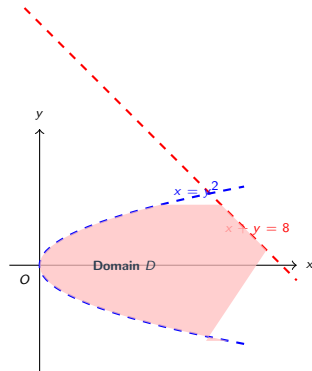
### Algebraic Conditions

We have two simultaneous conditions:

- ①  $\frac{1}{\sqrt{x - y^2}} : x - y^2 > 0 \iff x > y^2$   
(Region strictly to the right of parabola  $x = y^2$ )
- ②  $\ln(8 - x - y) : 8 - x - y > 0 \iff x + y < 8$   
(Region strictly below line  $x + y = 8$ )

**Domain Set:**

$$D = \{(x, y) \in \mathbb{R}^2 \mid x > y^2 \text{ and } x + y < 8\}.$$



# Domain Example 2: Linear Inequality (Slide 5)

## Example (MA1521 Slide 5)

Find and sketch the domain of  $f(x, y) = \ln(10 - 2x - 3y)$ .

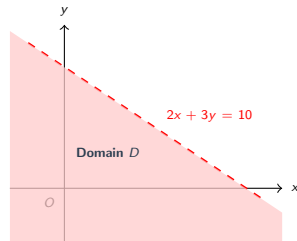
### Algebraic Condition

The natural logarithm requires a strictly positive argument:

$$\ln(10 - 2x - 3y) : 10 - 2x - 3y > 0 \iff 2x + 3y < 10.$$

- **Boundary line:**  $2x + 3y = 10$  (x-intercept 5, y-intercept  $10/3$ ).
- Inequality is strict ( $<$ ): Boundary is drawn as a **dashed line**.
- Testing  $(0, 0)$ :  $0 < 10$  (True  $\implies$  side containing origin).

**Domain Set:**  $D = \{(x, y) \in \mathbb{R}^2 \mid 2x + 3y < 10\}$ .



# Domain Example 3: Function of Three Variables (Slide 5)

## Example (MA1521 Slide 5)

Find and describe the domain of  $f(x, y, z) = \sqrt{9 - (x^2 + y^2 + z^2)}$ .

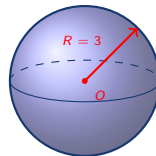
### Algebraic Condition in $\mathbb{R}^3$

The radicand of the square root must be non-negative:

$$9 - (x^2 + y^2 + z^2) \geq 0 \iff x^2 + y^2 + z^2 \leq 9 = 3^2.$$

- Boundary is the sphere  $x^2 + y^2 + z^2 = 9$  of radius  $R = 3$  centered at origin  $O(0, 0, 0)$ .
- The inequality ( $\leq$ ) includes the boundary sphere.

**Conclusion:** Domain  $D$  is the **solid ball** of center  $O$  and radius 3.



Solid Ball  $x^2 + y^2 + z^2 \leq 3^2$

# Graph of a Two-Variable Function (Slide 6)

## Definition (Graph in $\mathbb{R}^3$ )

The **graph** of function  $z = f(x, y)$  with domain  $D$  is:

$$\{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D \text{ and } z = f(x, y)\}.$$

It is a **surface** defined by the function.

## Question: How can we plot a surface?

- For  $s = g(t)$ , we pick points  $t_1, \dots, t_n$  and connect points with a smooth curve.
- For  $z = f(x, y)$ , this method fails. Instead:
  - For each  $z = z_0$ , draw the **level curve**  $z_0 = f(x, y)$ .
  - This is the **intersection** of the surface  $z = f(x, y)$  and the horizontal plane  $z = z_0$ :

$$L_{z_0} := \{(x, y) \mid f(x, y) = z_0\}.$$

# Level Curves Example 1: Linear Surface (Slide 7)

## Example (MA1521 Slide 7)

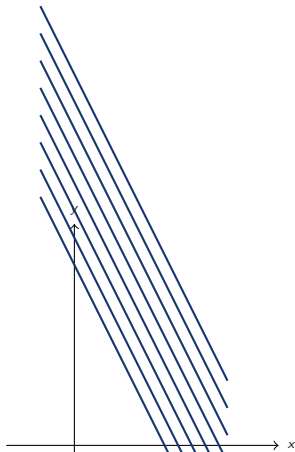
Draw the level curves of  $f(x, y) = 10 - 2x - y$ .

### Family of Level Curves

Set  $f(x, y) = k$ :

$$10 - 2x - y = k \iff 2x + y = 10 - k.$$

- For each  $k$ , this is a straight line with slope  $m = -2$ .
- For  $k = -5, -4, -3, -2, -1, 0, 1, 2, 3, 4$ :
- The level curves form a family of **parallel straight lines**.
- The 3D surface  $z = 10 - 2x - y$  is a **plane**.



# Level Curves Example 2: Circular Paraboloid (Slide 8)

## Example (MA1521 Slide 8)

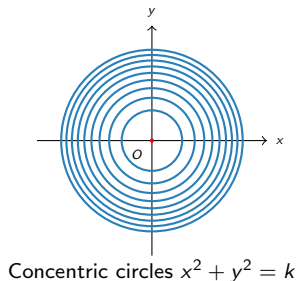
Draw the level curves of  $f(x, y) = x^2 + y^2$ .

### Family of Level Curves

Set  $f(x, y) = k$  ( $k \geq 0$ ):

$$x^2 + y^2 = k.$$

- $k = 0$ : Single point  $(0, 0)$  at apex/vertex.
- $k = 1, 2, 3, 4, 5, 6, 7, 8, 9$ :
- Concentric circles of radii  $r = \sqrt{k}$ .
- Circles become **closer together** as  $k$  grows  $\implies$  surface becomes **steeper**.
- The 3D surface is a **Circular Paraboloid**.



# Level Curves Example 3: Circular Cone (Slide 9)

## Example (MA1521 Slide 9)

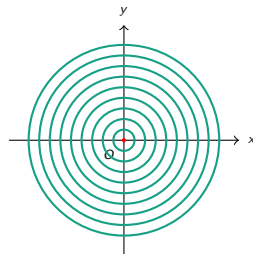
Draw the level curves of  $f(x, y) = \sqrt{x^2 + y^2}$ .

### Family of Level Curves

Set  $f(x, y) = k$  ( $k \geq 0$ ):

$$\sqrt{x^2 + y^2} = k \iff x^2 + y^2 = k^2.$$

- $k = 0$ : Single point  $(0, 0)$ .
- $k = 1, 2, 3, 4, 5, 6, 7, 8, 9$ :
- Circles centered at origin with radii  $r = k$ .
- Circles are **equally spaced** ( $\Delta r = 1$ )  $\implies$  surface has **constant slope**.
- The 3D surface is a **Circular Cone**.



Equidistant circles  $x^2 + y^2 = k^2$

# Level Curves Example 4: Hyperbolic Paraboloid (Slide 10)

## Example (MA1521 Slide 10)

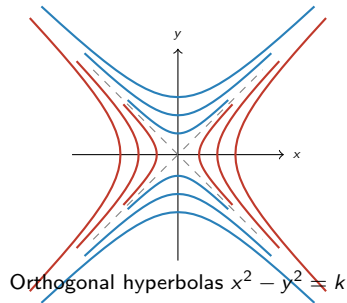
Draw the level curves of  $f(x, y) = x^2 - y^2$ .

### Family of Level Curves

Set  $f(x, y) = k$ :

$$x^2 - y^2 = k.$$

- $k = 0$ : Pair of intersecting lines  $y = \pm x$ .
- $k > 0$ : Hyperbolas opening along the  $x$ -axis.
- $k < 0$ : Hyperbolas opening along the  $y$ -axis.
- The 3D surface is a **Hyperbolic Paraboloid (Saddle Surface)**.
- Origin  $(0, 0)$  is a **saddle point**.



# Summary Checklist: Functions of Several Variables

## Part 1: Functions & Domains

- **Function**  $z = f(x, y)$ : Independent variables  $x, y \in D \subset \mathbb{R}^2$ , dependent variable  $z \in \mathbb{R}$ .
- **Natural Domain**: Largest subset of  $\mathbb{R}^2$  (or  $\mathbb{R}^n$ ) where  $f$  is well-defined.
- **Algebraic Rules**: Fractions  $Q \neq 0$ , Radicals  $\sqrt{g} \geq 0$ , Logs  $\ln(g) > 0$ .
- **Boundaries**:  $\geq, \leq \implies$  solid;  $>, < \implies$  dashed.

## Part 2: Graphs & Level Curves

- **Graph**: 3D surface  $z = f(x, y)$  in  $\mathbb{R}^3$ .
- **Level Curves**:  $L_{z_0} = \{(x, y) \mid f(x, y) = z_0\}$ .
- **Plane**:  $10 - 2x - y = k \implies$  parallel lines.
- **Paraboloid**:  $x^2 + y^2 = k \implies$  circles ( $r = \sqrt{k}$ ).
- **Cone**:  $\sqrt{x^2 + y^2} = k \implies$  circles ( $r = k$ ).
- **Saddle**:  $x^2 - y^2 = k \implies$  hyperbolas.

**Next Topic (Slide 11):** *Partial Derivatives*  $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$  *and Gradient Vector*  $\nabla f$ !

# Thank You!

Questions & Discussion

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